Fan Anomaly Detection

Machine Learning Final Project Report

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Data Source: DCASE 2024 Task 2 / MIMII DUE - fan subset. 1,000 normal training clips + 200 test clips (100 normal, 100 anomaly), 16 kHz mono, 10 s.

Approach: After downloading the dataset and creating the dataset card and building the audio preprocessing pipeline with reproducible parameters we moved to building the model training it and evaluating its performance through the ROC AUC, pAUC and average precision metrics.

Over 15+ iterations, approaches included LOF on handcrafted features, autoencoder variants, classifier embeddings, multiple transformer families (MIMI, AST, Wav2Vec2, CLAP, EnCodec), and various fusion strategies. **The final champion combined MIMI + AST + handcrafted features in a meta-stack with pseudo-anomaly training.**

The loss function of this chosen model : **logistic-regression loss (cross-entropy)**.

$$L = -\frac{1}{N}\sum_{i=1}^{N}[y_i \log(\hat{y}_i) + (1 - y_i)\log(1 - \hat{y}_i)]$$

Limitations: Performance converged near 0.71–0.73 AUC across all iterations and the key limiting factors were

- The absence of labeled anomalies at training time and if present the imbalanced nature of the data will create huge noise and a major setback in performance.
- source-to-target domain shift
- computing budget constraints that prevented full fine-tuning of large transformers.

Summary of the final method: The pipeline uses four acoustic feature sets derived from the audio clip (codec, transformer, self-supervised, and engineered), evaluates the anomaly separately using LOF scoring, and rejects those that fall below a minimum threshold of AUC value. The remaining anomalies are then normalized to a standard scale before undergoing ensemble learning using a meta-stacker based on logistic regression for synthetic anomaly training. Anomalies are detected by a thresholding process, and the faulty type is diagnosed using harmonic decomposition through an ODE neural network.

Results

- **Pooled AUC (vs DCASE baseline 0.610) : 0.7255**
- **Average Precision : 0.7312**
- Precision / Recall @p80 threshold : 0.640 / 0.730
- Confusion Matrix (100N + 100A): TN=59, FP=41, FN=27, TP=73
- Meta-stack weights HC +4.32 MIMI +3.00 AST +1.87 (bias −7.03)